

RAMNICU VALCEA, Romania

WMO: 153460

Lat: 45.0889N

Lon: 24.3628E

Elev: 782

StdP: 14.29

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.3	16.9	6.4	7.9	18.0	10.9	9.9	21.6	13.8	38.2	11.1	36.7	1.9	350	0.430

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.6	92.1	69.9	88.9	69.2	86.0	68.3	72.6	86.4	71.0	84.4	69.6	82.5	4.1	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
68.3	107.0	78.1	66.6	100.9	76.6	65.2	96.0	75.1	36.7	86.4	35.3	84.6	34.1	82.4	81.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.9	9.7	8.3	4.6	95.8	5.8	3.9	0.4	98.6	-3.0	100.8	-6.3	103.0	-10.6	105.8		
(5)			4.2	75.7	5.8	2.0	0.0	77.1	-3.4	78.2	-6.7	79.3	-10.9	80.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	53.4	31.9	36.1	44.1	54.2	62.9	69.4	73.0	72.9	63.9	53.7	43.8	34.4		
	DBStd	16.03	7.89	8.47	7.66	6.58	5.86	5.47	4.99	5.35	6.50	7.16	7.48	7.46		
	HDD50	1939	561	392	210	33	1	0	0	0	0	45	212	485		
	HDD65	4987	1026	810	649	328	109	22	2	6	98	353	635	949		
	CDD50	3195	0	2	27	158	401	581	712	710	416	161	27	0		
	CDD65	769	0	0	0	3	44	153	250	251	64	4	0	0		
	CDH74	7131	0	0	1	66	441	1391	2353	2250	578	51	0	0		
	CDH80	2587	0	0	0	8	76	456	942	937	164	4	0	0		
	WSAvg	3.6	2.9	3.4	4.1	4.2	4.1	4.0	4.0	3.9	3.7	3.3	3.0	2.9		
	PrecAvg	28.8	1.5	1.3	1.5	2.4	3.3	3.5	3.4	3.1	2.5	2.2	1.9	1.9		
(15) Precipitation	PrecMax	46.7	3.6	2.9	3.5	6.3	7.5	6.3	8.9	8.5	5.8	5.6	5.4	4.8		
	PrecMin	15.8	0.2	0.1	0.1	0.2	1.2	0.4	0.9	0.4	0.2	0.0	0.1	0.0		
	PrecStd	6.9	0.9	0.7	0.9	1.2	1.6	1.6	1.9	1.9	1.6	1.4	1.2	1.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.8	62.9	71.2	81.1	85.5	92.7	97.5	96.5	89.8	79.4	68.4	55.8		
		MCWB	45.6	49.8	53.6	60.0	64.4	71.5	69.0	69.6	66.7	62.0	55.8	47.4		
	2%	DB	50.1	56.9	66.3	74.8	82.1	88.5	92.1	92.1	84.9	74.2	62.8	50.7		
		MCWB	43.5	46.7	50.9	57.9	63.7	69.7	69.9	69.7	65.1	60.3	53.9	45.5		
	5%	DB	45.7	52.1	61.8	70.3	79.1	85.1	88.6	88.8	80.5	69.8	58.3	47.1		
		MCWB	41.3	44.1	48.9	55.3	62.7	68.5	69.6	69.1	63.6	58.5	51.3	43.3		
	10%	DB	42.0	47.8	57.4	66.7	75.7	82.0	85.4	85.4	76.6	65.5	54.6	44.3		
		MCWB	38.6	42.1	46.7	53.6	61.5	67.2	68.7	68.2	62.4	56.3	50.1	41.3		
	0.4%	WB	47.6	50.9	54.7	61.6	68.1	73.6	74.2	75.3	69.0	64.9	57.6	49.3		
		MCDB	53.3	61.0	67.4	76.3	80.6	87.9	88.1	87.7	83.8	75.3	64.9	53.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	44.7	47.7	52.0	59.0	65.6	71.4	72.4	72.7	66.6	61.7	55.0	46.5		
		MCDB	48.9	55.4	63.1	72.5	78.0	84.9	86.1	86.5	80.2	70.5	60.6	49.6		
	5%	WB	41.8	45.1	50.0	56.7	64.0	69.8	71.1	70.6	65.0	59.5	52.7	44.0		
		MCDB	45.0	50.4	60.1	67.6	76.1	82.5	84.4	84.2	76.8	67.6	57.0	46.5		
	10%	WB	39.1	42.7	47.8	54.8	62.3	68.1	69.7	69.0	63.4	57.4	50.5	41.8		
		MCDB	41.7	47.2	56.0	64.1	72.9	79.7	82.3	82.0	73.5	64.2	54.3	44.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	13.4	15.3	18.2	19.3	20.4	20.8	21.6	21.5	20.0	18.1	14.5	12.7		
		MCDBR	19.8	23.0	26.5	26.1	25.3	25.6	26.2	26.4	26.3	24.9	21.3	17.9		
		MCWBR	13.0	13.8	13.8	12.0	10.9	10.7	9.9	9.3	10.3	12.5	12.4	12.1		
		MCDBR	17.8	21.0	23.6	23.2	23.0	23.5	23.3	23.4	23.0	22.4	18.4	15.8		
	5% WB	MCWBR	12.0	13.1	12.8	11.2	10.6	10.4	10.0	9.8	9.9	11.8	11.0	11.2		
(40) Clear-Sky Solar Irradiance	taub		0.325	0.359	0.400	0.463	0.442	0.443	0.458	0.477	0.428	0.387	0.360	0.327		
	taud		2.310	2.227	2.183	2.035	2.166	2.205	2.172	2.109	2.210	2.314	2.333	2.341		
	Ebn at Noon		244	256	261	253	264	263	257	246	249	243	228	230		
	Edh at Noon		26	34	40	51	46	44	46	47	39	31	25	23		
(44) All-Sky Solar Radiation	RadAvg		445	709	1068	1419	1731	1897	1915	1712	1247	841	483	364		
	RadStd		55	99	119	148	152	157	147	135	114	108	56	47		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+1.15	N/A	N/A	N/A	N/A	N/A	-205	-324	+245	+128
(47)	Regional (78 neighbors)	+1.03	N/A	N/A	+1.44	N/A	N/A	N/A	-290	+148	+70

Nomenclature: See separate page